SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Aashika S P	**Reg. No.:**	192521068
Date:	9.10.2026	**Duration:**	60 minutes

Code of Conduct

I Aashika S P (192521068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Aashika S P

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

HONEY CHAIN: A BLOCKCHAIN-BASED SYSTEM FOR HONEY TRACEABILITY AND SMART BEEKEEPING MANAGEMENT

Project Vision and Stakeholder Identification

1.1 Introduction

Honey Chain is a blockchain-based software system designed to improve honey traceability, quality assurance, and beekeeping management. Traditional honey supply chains often face problems such as product adulteration, lack of transparency, difficulty in verifying honey origin, and inaccurate record maintenance.

The proposed system uses blockchain technology to maintain secure and tamper-resistant records of honey production, processing, packaging, and distribution. It also integrates smart beekeeping management features to help beekeepers monitor hive conditions, record honey production, and manage their apiaries efficiently.

Consumers can scan a QR code on honey packaging to view information about the honey's origin, harvesting date, processing details, and supply-chain history. The system improves consumer trust, supports beekeepers, and promotes transparency throughout the honey supply chain.

1.2 Project Vision

To develop a secure, transparent, and reliable blockchain-based platform that enables end-to-end honey traceability, smart beekeeping management, and product authenticity verification.

1.3 Project Objectives

- *To maintain secure and transparent records of honey production and distribution using blockchain technology.*
- *To track honey from the beehive to the final consumer.*
- *To help consumers verify honey authenticity through QR codes.*
- *To enable beekeepers to record hive information and honey harvesting details.*
- *To improve inventory management and reduce record-keeping errors.*
- *To provide reliable product information to consumers and distributors.*
- *To improve accountability and trust among all stakeholders.*

1.4 Stakeholder Identification

*1. **Beekeepers:** Record hive information, monitor hive conditions, and update honey harvesting details.*

*2. **Honey Producers:** Manage honey collection, processing, quality records, and packaging information.*

*3. **Suppliers and Distributors:** Record transportation, storage, and distribution activities.*

*4. **Consumers:** Scan QR codes to verify honey origin and view product history.*

*5. **Quality Control Inspectors:** Enter and verify laboratory test results and quality certifications.*

*6. **System Administrator:** Manage user accounts, permissions, and overall system operations.*

*7. **Retailers:** Maintain product inventory and access product traceability information.*

*8. **Development Team:** Design, develop, test, and maintain the Honey Chain application.*

*9. **Project Owner/Product Owner:** Define project requirements, prioritize the backlog, and approve completed features.*

2. Create Epics and User Stories

An epic is a large group of related requirements. User stories describe individual features from the perspective of the user.

Epic 1: User Registration and Authentication

***User Story 1.1:** As a beekeeper, I want to register and log in to the system so that I can securely manage my beekeeping information.*

***User Story 1.2:** As an administrator, I want to assign user roles so that each user can access only the features permitted for their role.*

Epic 2: Smart Beekeeping Management

***User Story 2.1:** As a beekeeper, I want to register my apiaries and beehives so that I can maintain organized hive records.*

***User Story 2.2:** As a beekeeper, I want to record hive conditions and honey production so that I can monitor hive performance.*

***User Story 2.3:** As a beekeeper, I want to record harvesting dates and quantities so that honey batches can be tracked accurately.*

Epic 3: Blockchain-Based Honey Traceability

***User Story 3.1:** As a honey producer, I want to create a unique batch record so that every honey batch can be identified throughout the supply chain.*

***User Story 3.2:** As a distributor, I want to record the transfer of honey batches so that their movement can be traced.*

***User Story 3.3:** As a system user, I want important traceability records to be stored on a blockchain so that unauthorized changes can be detected.*

Epic 4: Product Authentication and Quality Management

***User Story 4.1:** As a quality inspector, I want to record laboratory test results so that honey quality information can be verified.*

***User Story 4.2:** As a consumer, I want to scan a QR code so that I can view the origin and history of a honey product.*

User Story 4.3: *As a consumer, I want to view quality certification details so that I can make informed purchasing decisions.*

Epic 5: Inventory and Supply-Chain Management

User Story 5.1: *As a producer, I want to update honey stock quantities so that available inventory can be monitored.*

User Story 5.2: *As a retailer, I want to view batch availability so that I can manage product sales and restocking.*

3. Define Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied before a user story is considered complete.

User Story 1.1: User Registration and Login

> ***Given*** *a new beekeeper has valid registration details,* ***when*** *the beekeeper submits the registration form,* ***then*** *the system creates an account.*
>
> ***Given*** *a registered user enters correct login credentials,* ***when*** *the user logs in,* ***then*** *the system grants access to the appropriate dashboard.*
>
> ***Given*** *a user enters an incorrect password,* ***when*** *login is attempted,* ***then*** *the system displays an error message without granting access.*

User Story 2.1: Register Apiaries and Beehives

> ***Given*** *a beekeeper is logged in,* ***when*** *valid hive details are submitted,* ***then*** *the system creates a unique hive record.*
>
> *Each hive record must contain a unique ID, location, and registration details.*
>
> *The system must prevent duplicate hive IDs.*

User Story 2.3: Record Honey Harvesting

> ***Given*** *a registered hive exists,* ***when*** *the beekeeper enters the harvesting date and quantity,* ***then*** *the system saves the harvest record.*
>
> *The system must associate each harvest record with the correct hive.*
>
> *The system must reject missing or invalid mandatory values.*

User Story 3.1: Create a Honey Batch

> ***Given*** *a valid harvest record exists,* ***when*** *the producer creates a honey batch,* ***then*** *the system generates a unique batch ID.*
>
> *The batch record must contain the source hive or apiary, harvest date, quantity, and producer details.*

The batch ID must be linked to the corresponding traceability record.

User Story 3.2: Record Batch Transfers

***Given** a valid honey batch exists, **when** an authorized distributor records a transfer, **then** the system records the sender, receiver, date, and batch ID.*

The system must preserve previous transfer records.

Unauthorized users must not be permitted to modify transfer information.

User Story 4.1: Record Quality Test Results

***Given** a valid honey batch exists, **when** an authorized quality inspector submits test results, **then** the system saves the results with the inspector's details and test date.*

The system must distinguish verified test results from results that are pending verification.

The system must not automatically label a batch as safe or certified without the required verification.

User Story 4.2: QR Code-Based Honey Verification

***Given** a honey batch has been registered, **when** the system generates its QR code, **then** the code links to the correct batch record.*

***Given** a consumer scans the QR code, **when** the product record is available, **then** the system displays the available origin, harvest, processing, and distribution information.*

If the batch ID is invalid, the system must display an appropriate message.

4. Prioritize the Product Backlog

The product backlog contains all features required for the Honey Chain project. The features are prioritized according to business value, customer needs, security, and development dependencies.

*Priority levels are classified as **High, Medium, and Low**.*

Priority	***Backlog Item***	***Business Value***
High	*User registration and login*	*Provides secure access*
High	*User registration and login*	*Establishes source records*
High	*Honey harvesting records*	*Captures production information*
High	*Unique honey batch creation*	*Enables batch-level traceability*
High	*Blockchain record integration*	*Supports tamper-resistant records*
High	*QR code generation and verification*	*Enables consumer access*
Medium	*Quality testing records*	*Supports quality verification*
Medium	*Supply-chain transfer tracking*	*Records product movement*
Medium	*Inventory management*	*Helps monitor stock*
Medium	*Hive condition monitoring*	*Supports beekeeping management*

Low	*Advanced analytics dashboard*	*Provides detailed performance insights*
Low	*Automated alerts and notifications*	*Improves operational convenience*

Backlog Prioritization Explanation

The highest-priority features establish the foundation for the application. User authentication, hive registration, harvesting records, and batch creation must be implemented before reliable traceability can be achieved.

Blockchain integration and QR code verification are also high-priority features because they provide the main value of the Honey Chain system. Inventory management, advanced analytics, and notifications can be developed in later sprints after the core system is operational.

5. Plan the Sprint Backlog

A sprint backlog is a set of selected product backlog items and the tasks required to complete them during a sprint.

5.1 Sprint Details

Sprint Number: *Sprint 1*

Duration: *2 weeks*

Sprint Team: *Product Owner, Scrum Master, frontend developer, backend developer, blockchain developer, and tester.*

Sprint Objective: *Develop the basic Honey Chain application with secure user access, hive registration, honey harvesting records, and initial honey batch creation.*

5.2 Selected Sprint Backlog

User Story	***Development Tasks***	***Responsible Role***
User registration and login	*Design registration and login screens; implement authentication; test login validation*	*Frontend and backend developers*
Apiary and hive registration	*Design hive form; create hive database; implement hive registration and validation*	*Backend developer*
Honey harvesting records	*Design harvesting form; save harvest date and quantity; link harvest to hive*	*Backend developer*
Honey batch creation	*Design harvesting form; save harvest date and quantity; link harvest to hive*	*Backend and blockchain developers*
Initial traceability integration	*Design harvesting form; save harvest date and quantity; link harvest to hive*	*Blockchain developer*

Testing and integration	*Design harvesting form; save harvest date and quantity;*	*Tester and development team*

5.3 Sprint Activities

Week 1: Planning and Core Development

Conduct sprint planning and finalize user stories.

Design the system architecture and database.

Develop user registration and login.

Implement hive registration and harvesting records.

Week 2: Batch Management and Testing

Implement unique honey batch creation.

Develop the initial blockchain integration.

Integrate frontend and backend components.

Test the selected features and resolve defects.

Conduct the sprint review and retrospective.

6. Estimate Effort Using Story Points

Story points estimate the relative effort, complexity, and uncertainty of a user story. The Fibonacci sequence (1, 2, 3, 5, 8, 13) is used for estimation.

The estimates below are proposed planning estimates for the first sprint. They can be revised by the development team during Planning Poker.

User Story	***Story Points***	***Justification***
User registration and login	*3*	*Requires basic forms, authentication, and validation*
Apiary and hive registration	*3*	*Requires database design and input validation*
Honey harvesting records	*5*	*Requires date and quantity validation and hive association*
Honey batch creation	*5*	*Requires blockchain setup, transaction handling, and integration testing*
Initial blockchain integration	*8*	*Requires blockchain setup, transaction handling, and integration testing*

Total	*24*	***Estimated total effort for selected stories***

Justification of Estimation

3 points: *Relatively simple features with limited dependencies.*

5 points: *Features involving multiple data relationships and additional validation.*

8 points: *A more complex feature involving blockchain integration and technical uncertainty.*

The total is 24 story points. This is a tentative sprint commitment and should be checked against the team's previous velocity and available working time. If the team cannot complete all selected stories, the initial blockchain integration can be reduced to a smaller, testable task or moved to the next sprint.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

To develop the foundational Honey Chain application that allows authorized users to register and manage beehives, record honey harvesting information, create unique honey batch records, and prepare those records for blockchain-based traceability.

7.2 Expected Deliverables

At the end of Sprint 1, the team expects to deliver:

A working user registration and login module.

A hive registration module with unique hive identification.

A honey harvesting form with validated date and quantity fields.

A honey batch management module with unique batch IDs.

A database containing hive, harvest, and batch records.

An initial blockchain integration prototype, subject to successful testing.

A set of test cases and a report of completed tests.

Sprint review feedback and a list of improvements for Sprint 2.

7.3 Definition of Done

A user story is considered complete when:

Its acceptance criteria have been satisfied.

The implementation has been integrated into the application.

Relevant functional and validation tests have passed.

Identified critical defects have been resolved.

The feature has been demonstrated during the sprint review.

The Product Owner has reviewed the feature against the agreed requirements.

7.4 Future Sprint Plan

Sprint 2: *Complete blockchain transaction integration, improve batch transfer tracking, and implement QR code generation.*

Sprint 3: *Implement quality inspection records, consumer product verification, and inventory management.*

Sprint 4: *Add hive condition monitoring, reporting, system security improvements, and end-to-end testing.*

Conclusion

Honey Chain provides a practical approach to improving honey traceability and smart beekeeping management through blockchain technology. By maintaining linked records of hives, harvesting, batches, and supply-chain activities, the system aims to improve transparency and accountability.

Agile sprint planning helps the development team divide the project into manageable tasks, prioritize important features, estimate effort, and deliver working software incrementally. The first sprint focuses on building the foundation required for secure and reliable honey traceability, while later sprints extend the system with blockchain verification, QR codes, quality records, and smart management features.

The proposed system can help beekeepers maintain accurate records, help businesses manage honey distribution, and enable consumers to access reliable product-origin information.